

Interface Control Document

Purpose:

Defines all physical and logical interfaces between boards, sensors, and the ground station.

Electrical Interface Table:

Interface	Signal	Voltage	Connector	Direction	Description
I ² C Bus 1	SDA/SCL	3.3 V	4-pin JST	Bi	MPU6050, BMP280
UART	TX/RX	3.3 V	4-pin JST	Bi	NEO-6M GPS
SPI	MISO/MOSI/CLK/CS	3.3 V	6-pin	Master	microSD
GPIO	D0–D7	3.3 V	Header	Out	LED, switch
Power Rail	5V/3.3V/GND	5/3.3 V	XT30	Supply	EPS → Backplane
Wi-Fi	2.4 GHz	N/A	Wireless	Bi	OBC ↔ GCS

Protocol Summary:

- I²C: standard mode 100 kHz, CRC-8 validation.
- UART: 9600 baud, NMEA (GPS).
- SPI: 8 MHz clock, full duplex.
- Wi-Fi Telemetry: WebSocket/MQTT JSON, topic: sat/<id>/hk.

Timing diagram:

